

take place at Union level through the AI Office, which should have the powers of a market surveillance authority within the meaning of Regulation (EU) 2019/1020 for this purpose. In all other cases, national market surveillance authorities remain responsible for the supervision of AI systems. However, for general-purpose AI systems that can be used directly by deployers for at least one purpose that is classified as high-risk, market surveillance authorities should cooperate with the AI Office to carry out evaluations of compliance and inform the Board and other market surveillance authorities accordingly. Furthermore, market surveillance authorities should be able to request assistance from the AI Office where the market surveillance authority is unable to conclude an investigation on a high-risk AI system because of its inability to access certain information related to the general-purpose AI model on which the high-risk AI system is built. In such cases, the procedure regarding mutual assistance in cross-border cases in Chapter VI of Regulation (EU) 2019/1020 should apply *mutatis mutandis*.

- (162) To make best use of the centralised Union expertise and synergies at Union level, the powers of supervision and enforcement of the obligations on providers of general-purpose AI models should be a competence of the Commission. The AI Office should be able to carry out all necessary actions to monitor the effective implementation of this Regulation as regards general-purpose AI models. It should be able to investigate possible infringements of the rules on providers of general-purpose AI models both on its own initiative, following the results of its monitoring activities, or upon request from market surveillance authorities in line with the conditions set out in this Regulation. To support effective monitoring of the AI Office, it should provide for the possibility that downstream providers lodge complaints about possible infringements of the rules on providers of general-purpose AI models and systems.
- (163) With a view to complementing the governance systems for general-purpose AI models, the scientific panel should support the monitoring activities of the AI Office and may, in certain cases, provide qualified alerts to the AI Office which trigger follow-ups, such as investigations. This should be the case where the scientific panel has reason to suspect that a general-purpose AI model poses a concrete and identifiable risk at Union level. Furthermore, this should be the case where the scientific panel has reason to suspect that a general-purpose AI model meets the criteria that would lead to a classification as general-purpose AI model with systemic risk. To equip the scientific panel with the information necessary for the performance of those tasks, there should be a mechanism whereby the scientific panel can request the Commission to require documentation or information from a provider.
- (164) The AI Office should be able to take the necessary actions to monitor the effective implementation of and compliance with the obligations for providers of general-purpose AI models laid down in this Regulation. The AI Office should be able to investigate possible infringements in accordance with the powers provided for in this Regulation, including by requesting documentation and information, by conducting evaluations, as well as by requesting measures from providers of general-purpose AI models. When conducting evaluations, in order to make use of independent expertise, the AI Office should be able to involve independent experts to carry out the evaluations on its behalf. Compliance with the obligations should be enforceable, *inter alia*, through requests to take appropriate measures, including risk mitigation measures in the case of identified systemic risks as well as restricting the making available on the market, withdrawing or recalling the model. As a safeguard, where needed beyond the procedural rights provided for in this Regulation, providers of general-purpose AI models should have the procedural rights provided for in Article 18 of Regulation (EU) 2019/1020, which should apply *mutatis mutandis*, without prejudice to more specific procedural rights provided for by this Regulation.
- (165) The development of AI systems other than high-risk AI systems in accordance with the requirements of this Regulation may lead to a larger uptake of ethical and trustworthy AI in the Union. Providers of AI systems that are not high-risk should be encouraged to create codes of conduct, including related governance mechanisms, intended to foster the voluntary application of some or all of the mandatory requirements applicable to high-risk AI systems, adapted in light of the intended purpose of the systems and the lower risk involved and taking into account the available technical solutions and industry best practices such as model and data cards. Providers and, as appropriate, deployers of all AI systems, high-risk or not, and AI models should also be encouraged to apply on a voluntary basis additional requirements related, for example, to the elements of the Union's Ethics Guidelines for Trustworthy AI,

environmental sustainability, AI literacy measures, inclusive and diverse design and development of AI systems, including attention to vulnerable persons and accessibility to persons with disability, stakeholders' participation with the involvement, as appropriate, of relevant stakeholders such as business and civil society organisations, academia, research organisations, trade unions and consumer protection organisations in the design and development of AI systems, and diversity of the development teams, including gender balance. To ensure that the voluntary codes of conduct are effective, they should be based on clear objectives and key performance indicators to measure the achievement of those objectives. They should also be developed in an inclusive way, as appropriate, with the involvement of relevant stakeholders such as business and civil society organisations, academia, research organisations, trade unions and consumer protection organisation. The Commission may develop initiatives, including of a sectoral nature, to facilitate the lowering of technical barriers hindering cross-border exchange of data for AI development, including on data access infrastructure, semantic and technical interoperability of different types of data.

- (166) It is important that AI systems related to products that are not high-risk in accordance with this Regulation and thus are not required to comply with the requirements set out for high-risk AI systems are nevertheless safe when placed on the market or put into service. To contribute to this objective, Regulation (EU) 2023/988 of the European Parliament and of the Council ⁽⁵³⁾ would apply as a safety net.
- (167) In order to ensure trustful and constructive cooperation of competent authorities on Union and national level, all parties involved in the application of this Regulation should respect the confidentiality of information and data obtained in carrying out their tasks, in accordance with Union or national law. They should carry out their tasks and activities in such a manner as to protect, in particular, intellectual property rights, confidential business information and trade secrets, the effective implementation of this Regulation, public and national security interests, the integrity of criminal and administrative proceedings, and the integrity of classified information.
- (168) Compliance with this Regulation should be enforceable by means of the imposition of penalties and other enforcement measures. Member States should take all necessary measures to ensure that the provisions of this Regulation are implemented, including by laying down effective, proportionate and dissuasive penalties for their infringement, and to respect the *ne bis in idem* principle. In order to strengthen and harmonise administrative penalties for infringement of this Regulation, the upper limits for setting the administrative fines for certain specific infringements should be laid down. When assessing the amount of the fines, Member States should, in each individual case, take into account all relevant circumstances of the specific situation, with due regard in particular to the nature, gravity and duration of the infringement and of its consequences and to the size of the provider, in particular if the provider is an SME, including a start-up. The European Data Protection Supervisor should have the power to impose fines on Union institutions, agencies and bodies falling within the scope of this Regulation.
- (169) Compliance with the obligations on providers of general-purpose AI models imposed under this Regulation should be enforceable, inter alia, by means of fines. To that end, appropriate levels of fines should also be laid down for infringement of those obligations, including the failure to comply with measures requested by the Commission in accordance with this Regulation, subject to appropriate limitation periods in accordance with the principle of proportionality. All decisions taken by the Commission under this Regulation are subject to review by the Court of Justice of the European Union in accordance with the TFEU, including the unlimited jurisdiction of the Court of Justice with regard to penalties pursuant to Article 261 TFEU.
- (170) Union and national law already provide effective remedies to natural and legal persons whose rights and freedoms are adversely affected by the use of AI systems. Without prejudice to those remedies, any natural or legal person that has grounds to consider that there has been an infringement of this Regulation should be entitled to lodge a complaint to the relevant market surveillance authority.
- (171) Affected persons should have the right to obtain an explanation where a deployer's decision is based mainly upon the output from certain high-risk AI systems that fall within the scope of this Regulation and where that decision produces legal effects or similarly significantly affects those persons in a way that they consider to have an adverse

⁽⁵³⁾ Regulation (EU) 2023/988 of the European Parliament and of the Council of 10 May 2023 on general product safety, amending Regulation (EU) No 1025/2012 of the European Parliament and of the Council and Directive (EU) 2020/1828 of the European Parliament and the Council, and repealing Directive 2001/95/EC of the European Parliament and of the Council and Council Directive 87/357/EEC (OJ L 135, 23.5.2023, p. 1).

impact on their health, safety or fundamental rights. That explanation should be clear and meaningful and should provide a basis on which the affected persons are able to exercise their rights. The right to obtain an explanation should not apply to the use of AI systems for which exceptions or restrictions follow from Union or national law and should apply only to the extent this right is not already provided for under Union law.

- (172) Persons acting as whistleblowers on the infringements of this Regulation should be protected under the Union law. Directive (EU) 2019/1937 of the European Parliament and of the Council ⁽⁵⁴⁾ should therefore apply to the reporting of infringements of this Regulation and the protection of persons reporting such infringements.
- (173) In order to ensure that the regulatory framework can be adapted where necessary, the power to adopt acts in accordance with Article 290 TFEU should be delegated to the Commission to amend the conditions under which an AI system is not to be considered to be high-risk, the list of high-risk AI systems, the provisions regarding technical documentation, the content of the EU declaration of conformity the provisions regarding the conformity assessment procedures, the provisions establishing the high-risk AI systems to which the conformity assessment procedure based on assessment of the quality management system and assessment of the technical documentation should apply, the threshold, benchmarks and indicators, including by supplementing those benchmarks and indicators, in the rules for the classification of general-purpose AI models with systemic risk, the criteria for the designation of general-purpose AI models with systemic risk, the technical documentation for providers of general-purpose AI models and the transparency information for providers of general-purpose AI models. It is of particular importance that the Commission carry out appropriate consultations during its preparatory work, including at expert level, and that those consultations be conducted in accordance with the principles laid down in the Interinstitutional Agreement of 13 April 2016 on Better Law-Making ⁽⁵⁵⁾. In particular, to ensure equal participation in the preparation of delegated acts, the European Parliament and the Council receive all documents at the same time as Member States' experts, and their experts systematically have access to meetings of Commission expert groups dealing with the preparation of delegated acts.
- (174) Given the rapid technological developments and the technical expertise required to effectively apply this Regulation, the Commission should evaluate and review this Regulation by 2 August 2029 and every four years thereafter and report to the European Parliament and the Council. In addition, taking into account the implications for the scope of this Regulation, the Commission should carry out an assessment of the need to amend the list of high-risk AI systems and the list of prohibited practices once a year. Moreover, by 2 August 2028 and every four years thereafter, the Commission should evaluate and report to the European Parliament and to the Council on the need to amend the list of high-risk areas headings in the annex to this Regulation, the AI systems within the scope of the transparency obligations, the effectiveness of the supervision and governance system and the progress on the development of standardisation deliverables on energy efficient development of general-purpose AI models, including the need for further measures or actions. Finally, by 2 August 2028 and every three years thereafter, the Commission should evaluate the impact and effectiveness of voluntary codes of conduct to foster the application of the requirements provided for high-risk AI systems in the case of AI systems other than high-risk AI systems and possibly other additional requirements for such AI systems.
- (175) In order to ensure uniform conditions for the implementation of this Regulation, implementing powers should be conferred on the Commission. Those powers should be exercised in accordance with Regulation (EU) No 182/2011 of the European Parliament and of the Council ⁽⁵⁶⁾.
- (176) Since the objective of this Regulation, namely to improve the functioning of the internal market and to promote the uptake of human centric and trustworthy AI, while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection against harmful effects of AI systems in the Union and supporting innovation, cannot be sufficiently achieved by the Member States and can rather, by reason of the scale or effects of the action, be better achieved at Union level, the Union may adopt

⁽⁵⁴⁾ Directive (EU) 2019/1937 of the European Parliament and of the Council of 23 October 2019 on the protection of persons who report breaches of Union law (OJ L 305, 26.11.2019, p. 17).

⁽⁵⁵⁾ OJ L 123, 12.5.2016, p. 1.

⁽⁵⁶⁾ Regulation (EU) No 182/2011 of the European Parliament and of the Council of 16 February 2011 laying down the rules and general principles concerning mechanisms for control by Member States of the Commission's exercise of implementing powers (OJ L 55, 28.2.2011, p. 13).

measures in accordance with the principle of subsidiarity as set out in Article 5 TEU. In accordance with the principle of proportionality as set out in that Article, this Regulation does not go beyond what is necessary in order to achieve that objective.

- (177) In order to ensure legal certainty, ensure an appropriate adaptation period for operators and avoid disruption to the market, including by ensuring continuity of the use of AI systems, it is appropriate that this Regulation applies to the high-risk AI systems that have been placed on the market or put into service before the general date of application thereof, only if, from that date, those systems are subject to significant changes in their design or intended purpose. It is appropriate to clarify that, in this respect, the concept of significant change should be understood as equivalent in substance to the notion of substantial modification, which is used with regard only to high-risk AI systems pursuant to this Regulation. On an exceptional basis and in light of public accountability, operators of AI systems which are components of the large-scale IT systems established by the legal acts listed in an annex to this Regulation and operators of high-risk AI systems that are intended to be used by public authorities should, respectively, take the necessary steps to comply with the requirements of this Regulation by end of 2030 and by 2 August 2030.
- (178) Providers of high-risk AI systems are encouraged to start to comply, on a voluntary basis, with the relevant obligations of this Regulation already during the transitional period.
- (179) This Regulation should apply from 2 August 2026. However, taking into account the unacceptable risk associated with the use of AI in certain ways, the prohibitions as well as the general provisions of this Regulation should already apply from 2 February 2025. While the full effect of those prohibitions follows with the establishment of the governance and enforcement of this Regulation, anticipating the application of the prohibitions is important to take account of unacceptable risks and to have an effect on other procedures, such as in civil law. Moreover, the infrastructure related to the governance and the conformity assessment system should be operational before 2 August 2026, therefore the provisions on notified bodies and governance structure should apply from 2 August 2025. Given the rapid pace of technological advancements and adoption of general-purpose AI models, obligations for providers of general-purpose AI models should apply from 2 August 2025. Codes of practice should be ready by 2 May 2025 in view of enabling providers to demonstrate compliance on time. The AI Office should ensure that classification rules and procedures are up to date in light of technological developments. In addition, Member States should lay down and notify to the Commission the rules on penalties, including administrative fines, and ensure that they are properly and effectively implemented by the date of application of this Regulation. Therefore the provisions on penalties should apply from 2 August 2025.
- (180) The European Data Protection Supervisor and the European Data Protection Board were consulted in accordance with Article 42(1) and (2) of Regulation (EU) 2018/1725 and delivered their joint opinion on 18 June 2021,

HAVE ADOPTED THIS REGULATION:

CHAPTER I

GENERAL PROVISIONS

Article 1

Subject matter`

1. The purpose of this Regulation is to improve the functioning of the internal market and promote the uptake of human-centric and trustworthy artificial intelligence (AI), while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection, against the harmful effects of AI systems in the Union and supporting innovation.

2. This Regulation lays down:

- (a) harmonised rules for the placing on the market, the putting into service, and the use of AI systems in the Union;

- (b) prohibitions of certain AI practices;
- (c) specific requirements for high-risk AI systems and obligations for operators of such systems;
- (d) harmonised transparency rules for certain AI systems;
- (e) harmonised rules for the placing on the market of general-purpose AI models;
- (f) rules on market monitoring, market surveillance, governance and enforcement;
- (g) measures to support innovation, with a particular focus on SMEs, including start-ups.

Article 2

Scope

1. This Regulation applies to:

- (a) providers placing on the market or putting into service AI systems or placing on the market general-purpose AI models in the Union, irrespective of whether those providers are established or located within the Union or in a third country;
- (b) deployers of AI systems that have their place of establishment or are located within the Union;
- (c) providers and deployers of AI systems that have their place of establishment or are located in a third country, where the output produced by the AI system is used in the Union;
- (d) importers and distributors of AI systems;
- (e) product manufacturers placing on the market or putting into service an AI system together with their product and under their own name or trademark;
- (f) authorised representatives of providers, which are not established in the Union;
- (g) affected persons that are located in the Union.

2. For AI systems classified as high-risk AI systems in accordance with Article 6(1) related to products covered by the Union harmonisation legislation listed in Section B of Annex I, only Article 6(1), Articles 102 to 109 and Article 112 apply. Article 57 applies only in so far as the requirements for high-risk AI systems under this Regulation have been integrated in that Union harmonisation legislation.

3. This Regulation does not apply to areas outside the scope of Union law, and shall not, in any event, affect the competences of the Member States concerning national security, regardless of the type of entity entrusted by the Member States with carrying out tasks in relation to those competences.

This Regulation does not apply to AI systems where and in so far they are placed on the market, put into service, or used with or without modification exclusively for military, defence or national security purposes, regardless of the type of entity carrying out those activities.

This Regulation does not apply to AI systems which are not placed on the market or put into service in the Union, where the output is used in the Union exclusively for military, defence or national security purposes, regardless of the type of entity carrying out those activities.

4. This Regulation applies neither to public authorities in a third country nor to international organisations falling within the scope of this Regulation pursuant to paragraph 1, where those authorities or organisations use AI systems in the framework of international cooperation or agreements for law enforcement and judicial cooperation with the Union or with one or more Member States, provided that such a third country or international organisation provides adequate safeguards with respect to the protection of fundamental rights and freedoms of individuals.

5. This Regulation shall not affect the application of the provisions on the liability of providers of intermediary services as set out in Chapter II of Regulation (EU) 2022/2065.

6. This Regulation does not apply to AI systems or AI models, including their output, specifically developed and put into service for the sole purpose of scientific research and development.
7. Union law on the protection of personal data, privacy and the confidentiality of communications applies to personal data processed in connection with the rights and obligations laid down in this Regulation. This Regulation shall not affect Regulation (EU) 2016/679 or (EU) 2018/1725, or Directive 2002/58/EC or (EU) 2016/680, without prejudice to Article 10(5) and Article 59 of this Regulation.
8. This Regulation does not apply to any research, testing or development activity regarding AI systems or AI models prior to their being placed on the market or put into service. Such activities shall be conducted in accordance with applicable Union law. Testing in real world conditions shall not be covered by that exclusion.
9. This Regulation is without prejudice to the rules laid down by other Union legal acts related to consumer protection and product safety.
10. This Regulation does not apply to obligations of deployers who are natural persons using AI systems in the course of a purely personal non-professional activity.
11. This Regulation does not preclude the Union or Member States from maintaining or introducing laws, regulations or administrative provisions which are more favourable to workers in terms of protecting their rights in respect of the use of AI systems by employers, or from encouraging or allowing the application of collective agreements which are more favourable to workers.
12. This Regulation does not apply to AI systems released under free and open-source licences, unless they are placed on the market or put into service as high-risk AI systems or as an AI system that falls under Article 5 or 50.

Article 3

Definitions

For the purposes of this Regulation, the following definitions apply:

- (1) 'AI system' means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments;
- (2) 'risk' means the combination of the probability of an occurrence of harm and the severity of that harm;
- (3) 'provider' means a natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge;
- (4) 'deployer' means a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity;
- (5) 'authorised representative' means a natural or legal person located or established in the Union who has received and accepted a written mandate from a provider of an AI system or a general-purpose AI model to, respectively, perform and carry out on its behalf the obligations and procedures established by this Regulation;
- (6) 'importer' means a natural or legal person located or established in the Union that places on the market an AI system that bears the name or trademark of a natural or legal person established in a third country;
- (7) 'distributor' means a natural or legal person in the supply chain, other than the provider or the importer, that makes an AI system available on the Union market;
- (8) 'operator' means a provider, product manufacturer, deployer, authorised representative, importer or distributor;

- (9) 'placing on the market' means the first making available of an AI system or a general-purpose AI model on the Union market;
- (10) 'making available on the market' means the supply of an AI system or a general-purpose AI model for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge;
- (11) 'putting into service' means the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose;
- (12) 'intended purpose' means the use for which an AI system is intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation;
- (13) 'reasonably foreseeable misuse' means the use of an AI system in a way that is not in accordance with its intended purpose, but which may result from reasonably foreseeable human behaviour or interaction with other systems, including other AI systems;
- (14) 'safety component' means a component of a product or of an AI system which fulfils a safety function for that product or AI system, or the failure or malfunctioning of which endangers the health and safety of persons or property;
- (15) 'instructions for use' means the information provided by the provider to inform the deployer of, in particular, an AI system's intended purpose and proper use;
- (16) 'recall of an AI system' means any measure aiming to achieve the return to the provider or taking out of service or disabling the use of an AI system made available to deployers;
- (17) 'withdrawal of an AI system' means any measure aiming to prevent an AI system in the supply chain being made available on the market;
- (18) 'performance of an AI system' means the ability of an AI system to achieve its intended purpose;
- (19) 'notifying authority' means the national authority responsible for setting up and carrying out the necessary procedures for the assessment, designation and notification of conformity assessment bodies and for their monitoring;
- (20) 'conformity assessment' means the process of demonstrating whether the requirements set out in Chapter III, Section 2 relating to a high-risk AI system have been fulfilled;
- (21) 'conformity assessment body' means a body that performs third-party conformity assessment activities, including testing, certification and inspection;
- (22) 'notified body' means a conformity assessment body notified in accordance with this Regulation and other relevant Union harmonisation legislation;
- (23) 'substantial modification' means a change to an AI system after its placing on the market or putting into service which is not foreseen or planned in the initial conformity assessment carried out by the provider and as a result of which the compliance of the AI system with the requirements set out in Chapter III, Section 2 is affected or results in a modification to the intended purpose for which the AI system has been assessed;
- (24) 'CE marking' means a marking by which a provider indicates that an AI system is in conformity with the requirements set out in Chapter III, Section 2 and other applicable Union harmonisation legislation providing for its affixing;
- (25) 'post-market monitoring system' means all activities carried out by providers of AI systems to collect and review experience gained from the use of AI systems they place on the market or put into service for the purpose of identifying any need to immediately apply any necessary corrective or preventive actions;
- (26) 'market surveillance authority' means the national authority carrying out the activities and taking the measures pursuant to Regulation (EU) 2019/1020;

- (27) 'harmonised standard' means a harmonised standard as defined in Article 2(1), point (c), of Regulation (EU) No 1025/2012;
- (28) 'common specification' means a set of technical specifications as defined in Article 2, point (4) of Regulation (EU) No 1025/2012, providing means to comply with certain requirements established under this Regulation;
- (29) 'training data' means data used for training an AI system through fitting its learnable parameters;
- (30) 'validation data' means data used for providing an evaluation of the trained AI system and for tuning its non-learnable parameters and its learning process in order, inter alia, to prevent underfitting or overfitting;
- (31) 'validation data set' means a separate data set or part of the training data set, either as a fixed or variable split;
- (32) 'testing data' means data used for providing an independent evaluation of the AI system in order to confirm the expected performance of that system before its placing on the market or putting into service;
- (33) 'input data' means data provided to or directly acquired by an AI system on the basis of which the system produces an output;
- (34) 'biometric data' means personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data;
- (35) 'biometric identification' means the automated recognition of physical, physiological, behavioural, or psychological human features for the purpose of establishing the identity of a natural person by comparing biometric data of that individual to biometric data of individuals stored in a database;
- (36) 'biometric verification' means the automated, one-to-one verification, including authentication, of the identity of natural persons by comparing their biometric data to previously provided biometric data;
- (37) 'special categories of personal data' means the categories of personal data referred to in Article 9(1) of Regulation (EU) 2016/679, Article 10 of Directive (EU) 2016/680 and Article 10(1) of Regulation (EU) 2018/1725;
- (38) 'sensitive operational data' means operational data related to activities of prevention, detection, investigation or prosecution of criminal offences, the disclosure of which could jeopardise the integrity of criminal proceedings;
- (39) 'emotion recognition system' means an AI system for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data;
- (40) 'biometric categorisation system' means an AI system for the purpose of assigning natural persons to specific categories on the basis of their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons;
- (41) 'remote biometric identification system' means an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database;
- (42) 'real-time remote biometric identification system' means a remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention;
- (43) 'post-remote biometric identification system' means a remote biometric identification system other than a real-time remote biometric identification system;
- (44) 'publicly accessible space' means any publicly or privately owned physical place accessible to an undetermined number of natural persons, regardless of whether certain conditions for access may apply, and regardless of the potential capacity restrictions;

- (45) 'law enforcement authority' means:
- (a) any public authority competent for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security; or
 - (b) any other body or entity entrusted by Member State law to exercise public authority and public powers for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security;
- (46) 'law enforcement' means activities carried out by law enforcement authorities or on their behalf for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including safeguarding against and preventing threats to public security;
- (47) 'AI Office' means the Commission's function of contributing to the implementation, monitoring and supervision of AI systems and general-purpose AI models, and AI governance, provided for in Commission Decision of 24 January 2024; references in this Regulation to the AI Office shall be construed as references to the Commission;
- (48) 'national competent authority' means a notifying authority or a market surveillance authority; as regards AI systems put into service or used by Union institutions, agencies, offices and bodies, references to national competent authorities or market surveillance authorities in this Regulation shall be construed as references to the European Data Protection Supervisor;
- (49) 'serious incident' means an incident or malfunctioning of an AI system that directly or indirectly leads to any of the following:
- (a) the death of a person, or serious harm to a person's health;
 - (b) a serious and irreversible disruption of the management or operation of critical infrastructure;
 - (c) the infringement of obligations under Union law intended to protect fundamental rights;
 - (d) serious harm to property or the environment;
- (50) 'personal data' means personal data as defined in Article 4, point (1), of Regulation (EU) 2016/679;
- (51) 'non-personal data' means data other than personal data as defined in Article 4, point (1), of Regulation (EU) 2016/679;
- (52) 'profiling' means profiling as defined in Article 4, point (4), of Regulation (EU) 2016/679;
- (53) 'real-world testing plan' means a document that describes the objectives, methodology, geographical, population and temporal scope, monitoring, organisation and conduct of testing in real-world conditions;
- (54) 'sandbox plan' means a document agreed between the participating provider and the competent authority describing the objectives, conditions, timeframe, methodology and requirements for the activities carried out within the sandbox;
- (55) 'AI regulatory sandbox' means a controlled framework set up by a competent authority which offers providers or prospective providers of AI systems the possibility to develop, train, validate and test, where appropriate in real-world conditions, an innovative AI system, pursuant to a sandbox plan for a limited time under regulatory supervision;
- (56) 'AI literacy' means skills, knowledge and understanding that allow providers, deployers and affected persons, taking into account their respective rights and obligations in the context of this Regulation, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause;

- (57) ‘testing in real-world conditions’ means the temporary testing of an AI system for its intended purpose in real-world conditions outside a laboratory or otherwise simulated environment, with a view to gathering reliable and robust data and to assessing and verifying the conformity of the AI system with the requirements of this Regulation and it does not qualify as placing the AI system on the market or putting it into service within the meaning of this Regulation, provided that all the conditions laid down in Article 57 or 60 are fulfilled;
- (58) ‘subject’, for the purpose of real-world testing, means a natural person who participates in testing in real-world conditions;
- (59) ‘informed consent’ means a subject’s freely given, specific, unambiguous and voluntary expression of his or her willingness to participate in a particular testing in real-world conditions, after having been informed of all aspects of the testing that are relevant to the subject’s decision to participate;
- (60) ‘deep fake’ means AI-generated or manipulated image, audio or video content that resembles existing persons, objects, places, entities or events and would falsely appear to a person to be authentic or truthful;
- (61) ‘widespread infringement’ means any act or omission contrary to Union law protecting the interest of individuals, which:
- (a) has harmed or is likely to harm the collective interests of individuals residing in at least two Member States other than the Member State in which:
 - (i) the act or omission originated or took place;
 - (ii) the provider concerned, or, where applicable, its authorised representative is located or established; or
 - (iii) the deployer is established, when the infringement is committed by the deployer;
 - (b) has caused, causes or is likely to cause harm to the collective interests of individuals and has common features, including the same unlawful practice or the same interest being infringed, and is occurring concurrently, committed by the same operator, in at least three Member States;
- (62) ‘critical infrastructure’ means critical infrastructure as defined in Article 2, point (4), of Directive (EU) 2022/2557;
- (63) ‘general-purpose AI model’ means an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications, except AI models that are used for research, development or prototyping activities before they are placed on the market;
- (64) ‘high-impact capabilities’ means capabilities that match or exceed the capabilities recorded in the most advanced general-purpose AI models;
- (65) ‘systemic risk’ means a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain;
- (66) ‘general-purpose AI system’ means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;
- (67) ‘floating-point operation’ means any mathematical operation or assignment involving floating-point numbers, which are a subset of the real numbers typically represented on computers by an integer of fixed precision scaled by an integer exponent of a fixed base;
- (68) ‘downstream provider’ means a provider of an AI system, including a general-purpose AI system, which integrates an AI model, regardless of whether the AI model is provided by themselves and vertically integrated or provided by another entity based on contractual relations.

*Article 4***AI literacy**

Providers and deployers of AI systems shall take measures to ensure, to their best extent, a sufficient level of AI literacy of their staff and other persons dealing with the operation and use of AI systems on their behalf, taking into account their technical knowledge, experience, education and training and the context the AI systems are to be used in, and considering the persons or groups of persons on whom the AI systems are to be used.

CHAPTER II

PROHIBITED AI PRACTICES*Article 5***Prohibited AI practices**

1. The following AI practices shall be prohibited:

- (a) the placing on the market, the putting into service or the use of an AI system that deploys subliminal techniques beyond a person's consciousness or purposefully manipulative or deceptive techniques, with the objective, or the effect of materially distorting the behaviour of a person or a group of persons by appreciably impairing their ability to make an informed decision, thereby causing them to take a decision that they would not have otherwise taken in a manner that causes or is reasonably likely to cause that person, another person or group of persons significant harm;
- (b) the placing on the market, the putting into service or the use of an AI system that exploits any of the vulnerabilities of a natural person or a specific group of persons due to their age, disability or a specific social or economic situation, with the objective, or the effect, of materially distorting the behaviour of that person or a person belonging to that group in a manner that causes or is reasonably likely to cause that person or another person significant harm;
- (c) the placing on the market, the putting into service or the use of AI systems for the evaluation or classification of natural persons or groups of persons over a certain period of time based on their social behaviour or known, inferred or predicted personal or personality characteristics, with the social score leading to either or both of the following:
 - (i) detrimental or unfavourable treatment of certain natural persons or groups of persons in social contexts that are unrelated to the contexts in which the data was originally generated or collected;
 - (ii) detrimental or unfavourable treatment of certain natural persons or groups of persons that is unjustified or disproportionate to their social behaviour or its gravity;
- (d) the placing on the market, the putting into service for this specific purpose, or the use of an AI system for making risk assessments of natural persons in order to assess or predict the risk of a natural person committing a criminal offence, based solely on the profiling of a natural person or on assessing their personality traits and characteristics; this prohibition shall not apply to AI systems used to support the human assessment of the involvement of a person in a criminal activity, which is already based on objective and verifiable facts directly linked to a criminal activity;
- (e) the placing on the market, the putting into service for this specific purpose, or the use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage;
- (f) the placing on the market, the putting into service for this specific purpose, or the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons;

- (g) the placing on the market, the putting into service for this specific purpose, or the use of biometric categorisation systems that categorise individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation; this prohibition does not cover any labelling or filtering of lawfully acquired biometric datasets, such as images, based on biometric data or categorizing of biometric data in the area of law enforcement;
- (h) the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement, unless and in so far as such use is strictly necessary for one of the following objectives:
 - (i) the targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as the search for missing persons;
 - (ii) the prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack;
 - (iii) the localisation or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation or prosecution or executing a criminal penalty for offences referred to in Annex II and punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years.

Point (h) of the first subparagraph is without prejudice to Article 9 of Regulation (EU) 2016/679 for the processing of biometric data for purposes other than law enforcement.

2. The use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement for any of the objectives referred to in paragraph 1, first subparagraph, point (h), shall be deployed for the purposes set out in that point only to confirm the identity of the specifically targeted individual, and it shall take into account the following elements:

- (a) the nature of the situation giving rise to the possible use, in particular the seriousness, probability and scale of the harm that would be caused if the system were not used;
- (b) the consequences of the use of the system for the rights and freedoms of all persons concerned, in particular the seriousness, probability and scale of those consequences.

In addition, the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement for any of the objectives referred to in paragraph 1, first subparagraph, point (h), of this Article shall comply with necessary and proportionate safeguards and conditions in relation to the use in accordance with the national law authorising the use thereof, in particular as regards the temporal, geographic and personal limitations. The use of the 'real-time' remote biometric identification system in publicly accessible spaces shall be authorised only if the law enforcement authority has completed a fundamental rights impact assessment as provided for in Article 27 and has registered the system in the EU database according to Article 49. However, in duly justified cases of urgency, the use of such systems may be commenced without the registration in the EU database, provided that such registration is completed without undue delay.

3. For the purposes of paragraph 1, first subparagraph, point (h) and paragraph 2, each use for the purposes of law enforcement of a 'real-time' remote biometric identification system in publicly accessible spaces shall be subject to a prior authorisation granted by a judicial authority or an independent administrative authority whose decision is binding of the Member State in which the use is to take place, issued upon a reasoned request and in accordance with the detailed rules of national law referred to in paragraph 5. However, in a duly justified situation of urgency, the use of such system may be commenced without an authorisation provided that such authorisation is requested without undue delay, at the latest within 24 hours. If such authorisation is rejected, the use shall be stopped with immediate effect and all the data, as well as the results and outputs of that use shall be immediately discarded and deleted.

The competent judicial authority or an independent administrative authority whose decision is binding shall grant the authorisation only where it is satisfied, on the basis of objective evidence or clear indications presented to it, that the use of the 'real-time' remote biometric identification system concerned is necessary for, and proportionate to, achieving one of the

objectives specified in paragraph 1, first subparagraph, point (h), as identified in the request and, in particular, remains limited to what is strictly necessary concerning the period of time as well as the geographic and personal scope. In deciding on the request, that authority shall take into account the elements referred to in paragraph 2. No decision that produces an adverse legal effect on a person may be taken based solely on the output of the 'real-time' remote biometric identification system.

4. Without prejudice to paragraph 3, each use of a 'real-time' remote biometric identification system in publicly accessible spaces for law enforcement purposes shall be notified to the relevant market surveillance authority and the national data protection authority in accordance with the national rules referred to in paragraph 5. The notification shall, as a minimum, contain the information specified under paragraph 6 and shall not include sensitive operational data.

5. A Member State may decide to provide for the possibility to fully or partially authorise the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement within the limits and under the conditions listed in paragraph 1, first subparagraph, point (h), and paragraphs 2 and 3. Member States concerned shall lay down in their national law the necessary detailed rules for the request, issuance and exercise of, as well as supervision and reporting relating to, the authorisations referred to in paragraph 3. Those rules shall also specify in respect of which of the objectives listed in paragraph 1, first subparagraph, point (h), including which of the criminal offences referred to in point (h)(iii) thereof, the competent authorities may be authorised to use those systems for the purposes of law enforcement. Member States shall notify those rules to the Commission at the latest 30 days following the adoption thereof. Member States may introduce, in accordance with Union law, more restrictive laws on the use of remote biometric identification systems.

6. National market surveillance authorities and the national data protection authorities of Member States that have been notified of the use of 'real-time' remote biometric identification systems in publicly accessible spaces for law enforcement purposes pursuant to paragraph 4 shall submit to the Commission annual reports on such use. For that purpose, the Commission shall provide Member States and national market surveillance and data protection authorities with a template, including information on the number of the decisions taken by competent judicial authorities or an independent administrative authority whose decision is binding upon requests for authorisations in accordance with paragraph 3 and their result.

7. The Commission shall publish annual reports on the use of real-time remote biometric identification systems in publicly accessible spaces for law enforcement purposes, based on aggregated data in Member States on the basis of the annual reports referred to in paragraph 6. Those annual reports shall not include sensitive operational data of the related law enforcement activities.

8. This Article shall not affect the prohibitions that apply where an AI practice infringes other Union law.

CHAPTER III

HIGH-RISK AI SYSTEMS

SECTION 1

Classification of AI systems as high-risk

Article 6

Classification rules for high-risk AI systems

1. Irrespective of whether an AI system is placed on the market or put into service independently of the products referred to in points (a) and (b), that AI system shall be considered to be high-risk where both of the following conditions are fulfilled:

- (a) the AI system is intended to be used as a safety component of a product, or the AI system is itself a product, covered by the Union harmonisation legislation listed in Annex I;
- (b) the product whose safety component pursuant to point (a) is the AI system, or the AI system itself as a product, is required to undergo a third-party conformity assessment, with a view to the placing on the market or the putting into service of that product pursuant to the Union harmonisation legislation listed in Annex I.

2. In addition to the high-risk AI systems referred to in paragraph 1, AI systems referred to in Annex III shall be considered to be high-risk.

3. By derogation from paragraph 2, an AI system referred to in Annex III shall not be considered to be high-risk where it does not pose a significant risk of harm to the health, safety or fundamental rights of natural persons, including by not materially influencing the outcome of decision making.

The first subparagraph shall apply where any of the following conditions is fulfilled:

- (a) the AI system is intended to perform a narrow procedural task;
- (b) the AI system is intended to improve the result of a previously completed human activity;
- (c) the AI system is intended to detect decision-making patterns or deviations from prior decision-making patterns and is not meant to replace or influence the previously completed human assessment, without proper human review; or
- (d) the AI system is intended to perform a preparatory task to an assessment relevant for the purposes of the use cases listed in Annex III.

Notwithstanding the first subparagraph, an AI system referred to in Annex III shall always be considered to be high-risk where the AI system performs profiling of natural persons.

4. A provider who considers that an AI system referred to in Annex III is not high-risk shall document its assessment before that system is placed on the market or put into service. Such provider shall be subject to the registration obligation set out in Article 49(2). Upon request of national competent authorities, the provider shall provide the documentation of the assessment.

5. The Commission shall, after consulting the European Artificial Intelligence Board (the 'Board'), and no later than 2 February 2026, provide guidelines specifying the practical implementation of this Article in line with Article 96 together with a comprehensive list of practical examples of use cases of AI systems that are high-risk and not high-risk.

6. The Commission is empowered to adopt delegated acts in accordance with Article 97 in order to amend paragraph 3, second subparagraph, of this Article by adding new conditions to those laid down therein, or by modifying them, where there is concrete and reliable evidence of the existence of AI systems that fall under the scope of Annex III, but do not pose a significant risk of harm to the health, safety or fundamental rights of natural persons.

7. The Commission shall adopt delegated acts in accordance with Article 97 in order to amend paragraph 3, second subparagraph, of this Article by deleting any of the conditions laid down therein, where there is concrete and reliable evidence that this is necessary to maintain the level of protection of health, safety and fundamental rights provided for by this Regulation.

8. Any amendment to the conditions laid down in paragraph 3, second subparagraph, adopted in accordance with paragraphs 6 and 7 of this Article shall not decrease the overall level of protection of health, safety and fundamental rights provided for by this Regulation and shall ensure consistency with the delegated acts adopted pursuant to Article 7(1), and take account of market and technological developments.

Article 7

Amendments to Annex III

1. The Commission is empowered to adopt delegated acts in accordance with Article 97 to amend Annex III by adding or modifying use-cases of high-risk AI systems where both of the following conditions are fulfilled:

- (a) the AI systems are intended to be used in any of the areas listed in Annex III;
- (b) the AI systems pose a risk of harm to health and safety, or an adverse impact on fundamental rights, and that risk is equivalent to, or greater than, the risk of harm or of adverse impact posed by the high-risk AI systems already referred to in Annex III.

2. When assessing the condition under paragraph 1, point (b), the Commission shall take into account the following criteria:

- (a) the intended purpose of the AI system;
- (b) the extent to which an AI system has been used or is likely to be used;
- (c) the nature and amount of the data processed and used by the AI system, in particular whether special categories of personal data are processed;
- (d) the extent to which the AI system acts autonomously and the possibility for a human to override a decision or recommendations that may lead to potential harm;
- (e) the extent to which the use of an AI system has already caused harm to health and safety, has had an adverse impact on fundamental rights or has given rise to significant concerns in relation to the likelihood of such harm or adverse impact, as demonstrated, for example, by reports or documented allegations submitted to national competent authorities or by other reports, as appropriate;
- (f) the potential extent of such harm or such adverse impact, in particular in terms of its intensity and its ability to affect multiple persons or to disproportionately affect a particular group of persons;
- (g) the extent to which persons who are potentially harmed or suffer an adverse impact are dependent on the outcome produced with an AI system, in particular because for practical or legal reasons it is not reasonably possible to opt-out from that outcome;
- (h) the extent to which there is an imbalance of power, or the persons who are potentially harmed or suffer an adverse impact are in a vulnerable position in relation to the deployer of an AI system, in particular due to status, authority, knowledge, economic or social circumstances, or age;
- (i) the extent to which the outcome produced involving an AI system is easily corrigible or reversible, taking into account the technical solutions available to correct or reverse it, whereby outcomes having an adverse impact on health, safety or fundamental rights, shall not be considered to be easily corrigible or reversible;
- (j) the magnitude and likelihood of benefit of the deployment of the AI system for individuals, groups, or society at large, including possible improvements in product safety;
- (k) the extent to which existing Union law provides for:
 - (i) effective measures of redress in relation to the risks posed by an AI system, with the exclusion of claims for damages;
 - (ii) effective measures to prevent or substantially minimise those risks.

3. The Commission is empowered to adopt delegated acts in accordance with Article 97 to amend the list in Annex III by removing high-risk AI systems where both of the following conditions are fulfilled:

- (a) the high-risk AI system concerned no longer poses any significant risks to fundamental rights, health or safety, taking into account the criteria listed in paragraph 2;
- (b) the deletion does not decrease the overall level of protection of health, safety and fundamental rights under Union law.

SECTION 2

Requirements for high-risk AI systems

Article 8

Compliance with the requirements

1. High-risk AI systems shall comply with the requirements laid down in this Section, taking into account their intended purpose as well as the generally acknowledged state of the art on AI and AI-related technologies. The risk management system referred to in Article 9 shall be taken into account when ensuring compliance with those requirements.

2. Where a product contains an AI system, to which the requirements of this Regulation as well as requirements of the Union harmonisation legislation listed in Section A of Annex I apply, providers shall be responsible for ensuring that their product is fully compliant with all applicable requirements under applicable Union harmonisation legislation. In ensuring the compliance of high-risk AI systems referred to in paragraph 1 with the requirements set out in this Section, and in order to ensure consistency, avoid duplication and minimise additional burdens, providers shall have a choice of integrating, as appropriate, the necessary testing and reporting processes, information and documentation they provide with regard to their product into documentation and procedures that already exist and are required under the Union harmonisation legislation listed in Section A of Annex I.

Article 9

Risk management system

1. A risk management system shall be established, implemented, documented and maintained in relation to high-risk AI systems.

2. The risk management system shall be understood as a continuous iterative process planned and run throughout the entire lifecycle of a high-risk AI system, requiring regular systematic review and updating. It shall comprise the following steps:

- (a) the identification and analysis of the known and the reasonably foreseeable risks that the high-risk AI system can pose to health, safety or fundamental rights when the high-risk AI system is used in accordance with its intended purpose;
- (b) the estimation and evaluation of the risks that may emerge when the high-risk AI system is used in accordance with its intended purpose, and under conditions of reasonably foreseeable misuse;
- (c) the evaluation of other risks possibly arising, based on the analysis of data gathered from the post-market monitoring system referred to in Article 72;
- (d) the adoption of appropriate and targeted risk management measures designed to address the risks identified pursuant to point (a).

3. The risks referred to in this Article shall concern only those which may be reasonably mitigated or eliminated through the development or design of the high-risk AI system, or the provision of adequate technical information.

4. The risk management measures referred to in paragraph 2, point (d), shall give due consideration to the effects and possible interaction resulting from the combined application of the requirements set out in this Section, with a view to minimising risks more effectively while achieving an appropriate balance in implementing the measures to fulfil those requirements.

5. The risk management measures referred to in paragraph 2, point (d), shall be such that the relevant residual risk associated with each hazard, as well as the overall residual risk of the high-risk AI systems is judged to be acceptable.

In identifying the most appropriate risk management measures, the following shall be ensured:

- (a) elimination or reduction of risks identified and evaluated pursuant to paragraph 2 in as far as technically feasible through adequate design and development of the high-risk AI system;
- (b) where appropriate, implementation of adequate mitigation and control measures addressing risks that cannot be eliminated;
- (c) provision of information required pursuant to Article 13 and, where appropriate, training to deployers.

With a view to eliminating or reducing risks related to the use of the high-risk AI system, due consideration shall be given to the technical knowledge, experience, education, the training to be expected by the deployer, and the presumable context in which the system is intended to be used.

6. High-risk AI systems shall be tested for the purpose of identifying the most appropriate and targeted risk management measures. Testing shall ensure that high-risk AI systems perform consistently for their intended purpose and that they are in compliance with the requirements set out in this Section.
7. Testing procedures may include testing in real-world conditions in accordance with Article 60.
8. The testing of high-risk AI systems shall be performed, as appropriate, at any time throughout the development process, and, in any event, prior to their being placed on the market or put into service. Testing shall be carried out against prior defined metrics and probabilistic thresholds that are appropriate to the intended purpose of the high-risk AI system.
9. When implementing the risk management system as provided for in paragraphs 1 to 7, providers shall give consideration to whether in view of its intended purpose the high-risk AI system is likely to have an adverse impact on persons under the age of 18 and, as appropriate, other vulnerable groups.
10. For providers of high-risk AI systems that are subject to requirements regarding internal risk management processes under other relevant provisions of Union law, the aspects provided in paragraphs 1 to 9 may be part of, or combined with, the risk management procedures established pursuant to that law.

Article 10

Data and data governance

1. High-risk AI systems which make use of techniques involving the training of AI models with data shall be developed on the basis of training, validation and testing data sets that meet the quality criteria referred to in paragraphs 2 to 5 whenever such data sets are used.
2. Training, validation and testing data sets shall be subject to data governance and management practices appropriate for the intended purpose of the high-risk AI system. Those practices shall concern in particular:
 - (a) the relevant design choices;
 - (b) data collection processes and the origin of data, and in the case of personal data, the original purpose of the data collection;
 - (c) relevant data-preparation processing operations, such as annotation, labelling, cleaning, updating, enrichment and aggregation;
 - (d) the formulation of assumptions, in particular with respect to the information that the data are supposed to measure and represent;
 - (e) an assessment of the availability, quantity and suitability of the data sets that are needed;
 - (f) examination in view of possible biases that are likely to affect the health and safety of persons, have a negative impact on fundamental rights or lead to discrimination prohibited under Union law, especially where data outputs influence inputs for future operations;
 - (g) appropriate measures to detect, prevent and mitigate possible biases identified according to point (f);
 - (h) the identification of relevant data gaps or shortcomings that prevent compliance with this Regulation, and how those gaps and shortcomings can be addressed.
3. Training, validation and testing data sets shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose. They shall have the appropriate statistical properties, including, where applicable, as regards the persons or groups of persons in relation to whom the high-risk AI system is intended to be used. Those characteristics of the data sets may be met at the level of individual data sets or at the level of a combination thereof.
4. Data sets shall take into account, to the extent required by the intended purpose, the characteristics or elements that are particular to the specific geographical, contextual, behavioural or functional setting within which the high-risk AI system is intended to be used.

5. To the extent that it is strictly necessary for the purpose of ensuring bias detection and correction in relation to the high-risk AI systems in accordance with paragraph (2), points (f) and (g) of this Article, the providers of such systems may exceptionally process special categories of personal data, subject to appropriate safeguards for the fundamental rights and freedoms of natural persons. In addition to the provisions set out in Regulations (EU) 2016/679 and (EU) 2018/1725 and Directive (EU) 2016/680, all the following conditions must be met in order for such processing to occur:

- (a) the bias detection and correction cannot be effectively fulfilled by processing other data, including synthetic or anonymised data;
- (b) the special categories of personal data are subject to technical limitations on the re-use of the personal data, and state-of-the-art security and privacy-preserving measures, including pseudonymisation;
- (c) the special categories of personal data are subject to measures to ensure that the personal data processed are secured, protected, subject to suitable safeguards, including strict controls and documentation of the access, to avoid misuse and ensure that only authorised persons have access to those personal data with appropriate confidentiality obligations;
- (d) the special categories of personal data are not to be transmitted, transferred or otherwise accessed by other parties;
- (e) the special categories of personal data are deleted once the bias has been corrected or the personal data has reached the end of its retention period, whichever comes first;
- (f) the records of processing activities pursuant to Regulations (EU) 2016/679 and (EU) 2018/1725 and Directive (EU) 2016/680 include the reasons why the processing of special categories of personal data was strictly necessary to detect and correct biases, and why that objective could not be achieved by processing other data.

6. For the development of high-risk AI systems not using techniques involving the training of AI models, paragraphs 2 to 5 apply only to the testing data sets.

Article 11

Technical documentation

1. The technical documentation of a high-risk AI system shall be drawn up before that system is placed on the market or put into service and shall be kept up-to-date.

The technical documentation shall be drawn up in such a way as to demonstrate that the high-risk AI system complies with the requirements set out in this Section and to provide national competent authorities and notified bodies with the necessary information in a clear and comprehensive form to assess the compliance of the AI system with those requirements. It shall contain, at a minimum, the elements set out in Annex IV. SMEs, including start-ups, may provide the elements of the technical documentation specified in Annex IV in a simplified manner. To that end, the Commission shall establish a simplified technical documentation form targeted at the needs of small and microenterprises. Where an SME, including a start-up, opts to provide the information required in Annex IV in a simplified manner, it shall use the form referred to in this paragraph. Notified bodies shall accept the form for the purposes of the conformity assessment.

2. Where a high-risk AI system related to a product covered by the Union harmonisation legislation listed in Section A of Annex I is placed on the market or put into service, a single set of technical documentation shall be drawn up containing all the information set out in paragraph 1, as well as the information required under those legal acts.

3. The Commission is empowered to adopt delegated acts in accordance with Article 97 in order to amend Annex IV, where necessary, to ensure that, in light of technical progress, the technical documentation provides all the information necessary to assess the compliance of the system with the requirements set out in this Section.

*Article 12***Record-keeping**

1. High-risk AI systems shall technically allow for the automatic recording of events (logs) over the lifetime of the system.
2. In order to ensure a level of traceability of the functioning of a high-risk AI system that is appropriate to the intended purpose of the system, logging capabilities shall enable the recording of events relevant for:
 - (a) identifying situations that may result in the high-risk AI system presenting a risk within the meaning of Article 79(1) or in a substantial modification;
 - (b) facilitating the post-market monitoring referred to in Article 72; and
 - (c) monitoring the operation of high-risk AI systems referred to in Article 26(5).
3. For high-risk AI systems referred to in point 1 (a), of Annex III, the logging capabilities shall provide, at a minimum:
 - (a) recording of the period of each use of the system (start date and time and end date and time of each use);
 - (b) the reference database against which input data has been checked by the system;
 - (c) the input data for which the search has led to a match;
 - (d) the identification of the natural persons involved in the verification of the results, as referred to in Article 14(5).

*Article 13***Transparency and provision of information to deployers**

1. High-risk AI systems shall be designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately. An appropriate type and degree of transparency shall be ensured with a view to achieving compliance with the relevant obligations of the provider and deployer set out in Section 3.
2. High-risk AI systems shall be accompanied by instructions for use in an appropriate digital format or otherwise that include concise, complete, correct and clear information that is relevant, accessible and comprehensible to deployers.
3. The instructions for use shall contain at least the following information:
 - (a) the identity and the contact details of the provider and, where applicable, of its authorised representative;
 - (b) the characteristics, capabilities and limitations of performance of the high-risk AI system, including:
 - (i) its intended purpose;
 - (ii) the level of accuracy, including its metrics, robustness and cybersecurity referred to in Article 15 against which the high-risk AI system has been tested and validated and which can be expected, and any known and foreseeable circumstances that may have an impact on that expected level of accuracy, robustness and cybersecurity;
 - (iii) any known or foreseeable circumstance, related to the use of the high-risk AI system in accordance with its intended purpose or under conditions of reasonably foreseeable misuse, which may lead to risks to the health and safety or fundamental rights referred to in Article 9(2);
 - (iv) where applicable, the technical capabilities and characteristics of the high-risk AI system to provide information that is relevant to explain its output;

- (v) when appropriate, its performance regarding specific persons or groups of persons on which the system is intended to be used;
 - (vi) when appropriate, specifications for the input data, or any other relevant information in terms of the training, validation and testing data sets used, taking into account the intended purpose of the high-risk AI system;
 - (vii) where applicable, information to enable deployers to interpret the output of the high-risk AI system and use it appropriately;
- (c) the changes to the high-risk AI system and its performance which have been pre-determined by the provider at the moment of the initial conformity assessment, if any;
 - (d) the human oversight measures referred to in Article 14, including the technical measures put in place to facilitate the interpretation of the outputs of the high-risk AI systems by the deployers;
 - (e) the computational and hardware resources needed, the expected lifetime of the high-risk AI system and any necessary maintenance and care measures, including their frequency, to ensure the proper functioning of that AI system, including as regards software updates;
 - (f) where relevant, a description of the mechanisms included within the high-risk AI system that allows deployers to properly collect, store and interpret the logs in accordance with Article 12.

Article 14

Human oversight

1. High-risk AI systems shall be designed and developed in such a way, including with appropriate human-machine interface tools, that they can be effectively overseen by natural persons during the period in which they are in use.
2. Human oversight shall aim to prevent or minimise the risks to health, safety or fundamental rights that may emerge when a high-risk AI system is used in accordance with its intended purpose or under conditions of reasonably foreseeable misuse, in particular where such risks persist despite the application of other requirements set out in this Section.
3. The oversight measures shall be commensurate with the risks, level of autonomy and context of use of the high-risk AI system, and shall be ensured through either one or both of the following types of measures:
 - (a) measures identified and built, when technically feasible, into the high-risk AI system by the provider before it is placed on the market or put into service;
 - (b) measures identified by the provider before placing the high-risk AI system on the market or putting it into service and that are appropriate to be implemented by the deployer.
4. For the purpose of implementing paragraphs 1, 2 and 3, the high-risk AI system shall be provided to the deployer in such a way that natural persons to whom human oversight is assigned are enabled, as appropriate and proportionate:
 - (a) to properly understand the relevant capacities and limitations of the high-risk AI system and be able to duly monitor its operation, including in view of detecting and addressing anomalies, dysfunctions and unexpected performance;
 - (b) to remain aware of the possible tendency of automatically relying or over-relying on the output produced by a high-risk AI system (automation bias), in particular for high-risk AI systems used to provide information or recommendations for decisions to be taken by natural persons;
 - (c) to correctly interpret the high-risk AI system's output, taking into account, for example, the interpretation tools and methods available;